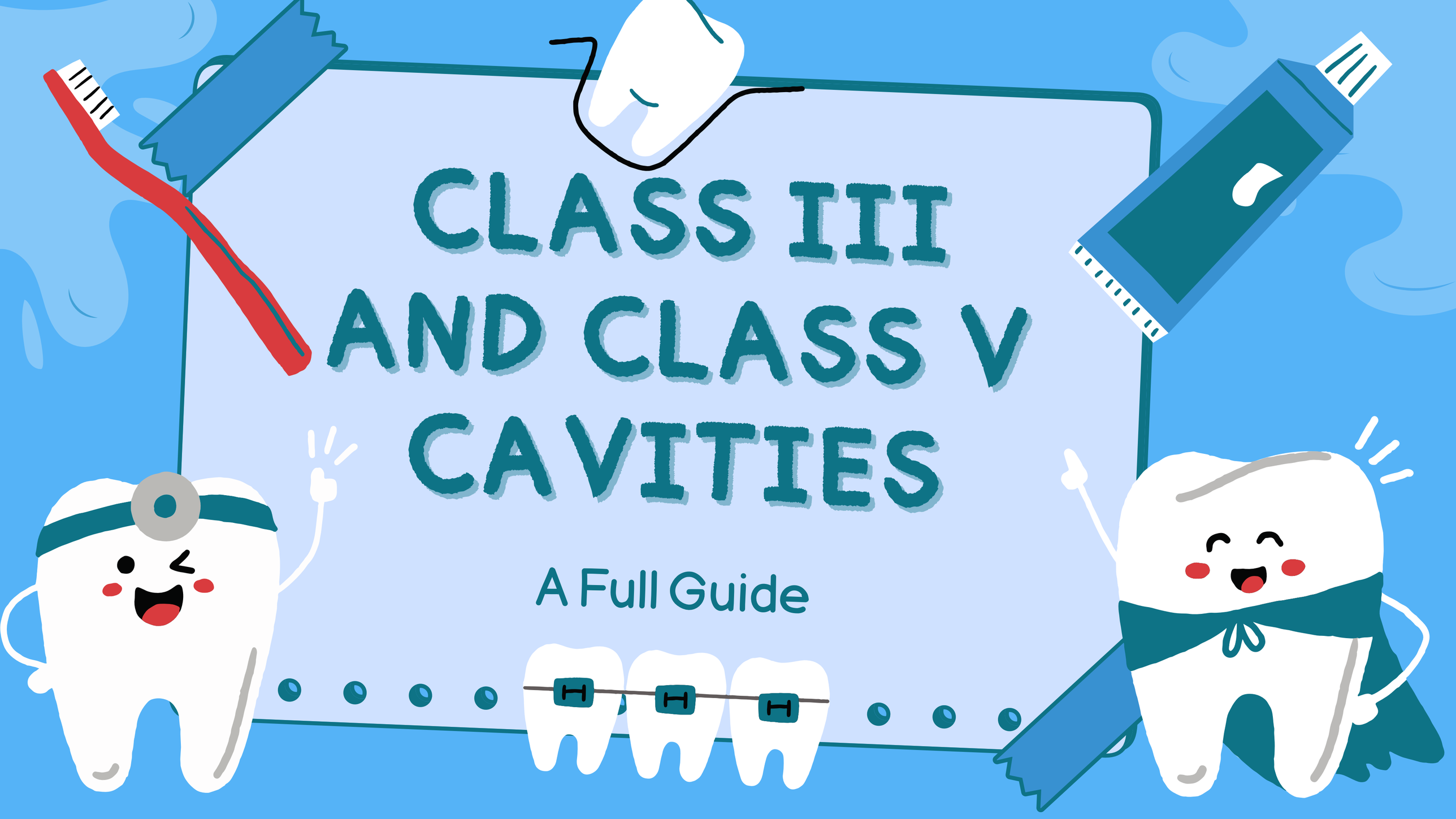
CLASS III AND CLASS V CAVITIES

A Full Guide





Introduction



Class III and V cavities are some of the most common type of cavities encountered in everyday dentistry.

Understanding their principles is essential before practical application.

Your goals while drilling your cavity should be:

- Every cut you make into the tooth should have a purpose.
- Establish correct outline form.
- Maintain the proper depth for each cavity type.
- Preserve sound tooth structure.
- Create walls and margins that support the future restoration.
- Preserve pulp vitality wherever possible.





What You Need for the Perfect Prep



Micromotor for low speed



High/low speed contra



Mirror + Explorer



Probe



330 carbide bur



Round carbide bur



Tapered with round end yellow

Class III Cavities

What is a Class III Cavity:

According to GV Black classification, Class III cavities are defined as carious lesions located on the proximal surfaces of incisors and canines that do not involve the incisal angle. They are usually located on the palatal/lingual surface of the teeth and can cross to the facial surface if they are deep enough.

Class III cavities are a smooth surface caries. This means that they are typically present in patients with poor oral hygiene and/or very high caries index.



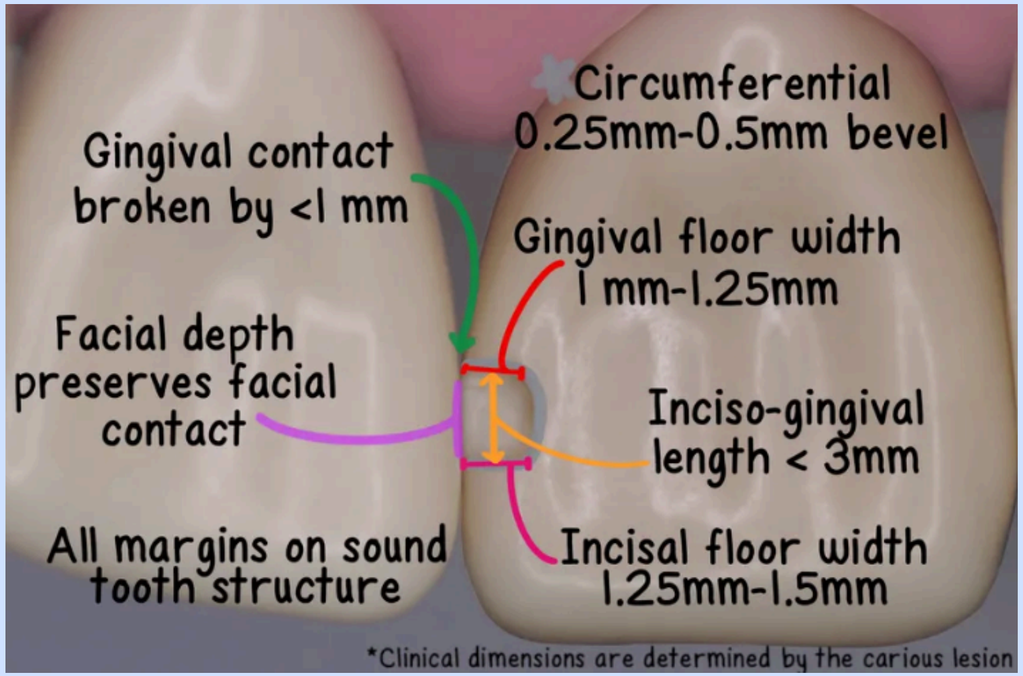


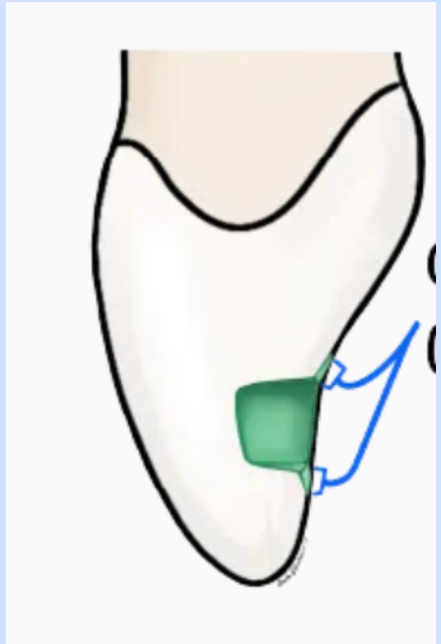
Class III Cavity Criteria

In a clinical setting, Class III cavity outline is usually determined by the **extent** and **pattern** of caries.

However, in preclinical setting, our acrylic tooth is completely sound with no caries pattern to follow.

That is why we follow specific criteria and measurements to ensure the correct outline form.





Class III Cavity Criteria

Lets understand the criteria simply using this video:



Class III Cavity Prep

Lets Prep a class III cavity on an Upper Central tooth :



Class III Cavity Restoration



As mentioned before, Class III cavities usually occur in patients with :

- Poor oral hygiene
- High caries index

So the dentist has to think carefully about the type of restoration to place.
Using Glass Ionomer restoration as an **intermediate restoration** till the conditions of the patient's oral hygiene improves is a very good option.
Glass ionomer restorations also release **fluoride** which could help **decrease the risk of recurrent caries (which is useful in high caries index patient)**

However, composite restoration can also be used in cases where the patient has improved their oral hygine or they have **high esthetic demands.**
So in the end, the dentist has to evaluate the situation in a **case-by- case basis** and then make their choice.





What You Need for the Perfect Restoration

Micromotor





Low speed contra



Composite

OR





Matrix band

Lightcure

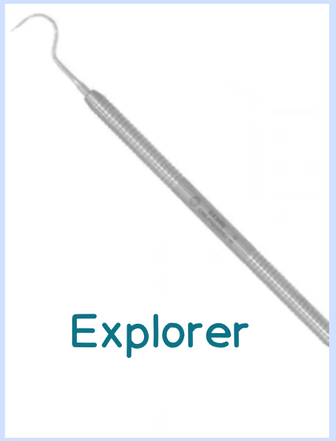




Plastic Instrument



Ball Burnisher



Explorer



Microbrush

Wedges





Polishing kit



Finishing Discs





Finishing Stones



Isolation

In a clinical setting, proper isolation is key to a succsessful restoration.

Composite restorations are **especially hydrophobic (ie sensitive to moisture).** So your working field should be isolated properly from moisture in the oral cavity like saliva and blood.

Isolation can be done using several methods depending on each case.

Rubber Dam isolation is considered the **GOLDEN STANDARD** of isolation.

However, in some cases, rubber dam application is not possible .

Alternatives to rubber dam can be :

- Cotton roll isolation
- Isolite system
- Combination of bite blocks, high volume suction, cheeck retractors.









Class III Cavity Restoration



Class V Cavities

What is a Class V Cavity?

According to GV Black classification, they are cavities located on the gingival third (cervical area) of the facial or lingual surfaces of any tooth, including both anterior and posterior teeth.

Class V cavities are a smooth surface caries. This means that they are typically present in patients with poor oral hygiene and/or very high caries index.





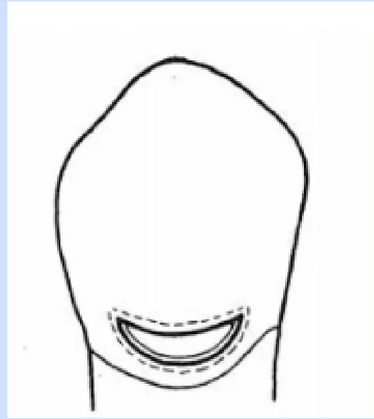
Class V Cavity Criteria

In a clinical setting, Class V cavity outline is usually determined by the extent and pattern of caries.

However, in preclinical setting, our acrylic tooth is completely sound with no caries pattern to follow.

That is why we follow specific criteria and measurements to ensure the correct outline form.

Criteria:



- Depth of the cavity should be 1.5 mm (equal to height of 330 bur)
- Width of the cavity should be 1.5 mm (equal to height of 330 bur)
- Smooth and even floor
- Defined, parallel and rounded walls
- No undermined enamel
- Correct outline form following the natural contour of the gingival margin.
- Correct location. It should be contained in the cervical third only not extended further.
- To make it easier imagine you are drawing a smiley face in the cervical third of the tooth, just like this(:D)



Class V Cavity Prep

Class V Cavity Restoration







As mentioned before, Class V cavities usually occur in patients with :

- Poor oral hygiene
- High caries index

So the dentist has to think carefully about the type of restoration to place.
Using Glass Ionomer restoration as an **intermediate restoration** till the conditions of the patient's oral hygiene improves is a very good option.
Glass ionomer restorations also release **fluoride** which could help **decrease the risk of recurrent caries (which is useful in high caries index patient)**

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What You Need for the Perfect Restoration

Micromotor





Low speed contra



Composite

OR





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Lightcure

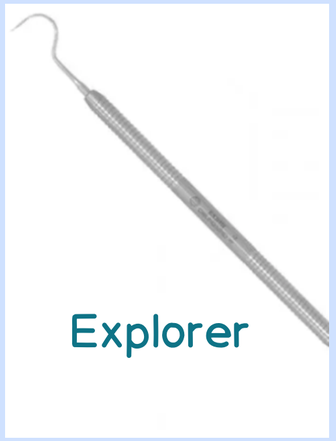




Plastic Instrument



Ball Burnisher



Explorer



Microbrush

Wedges





Polishing kit



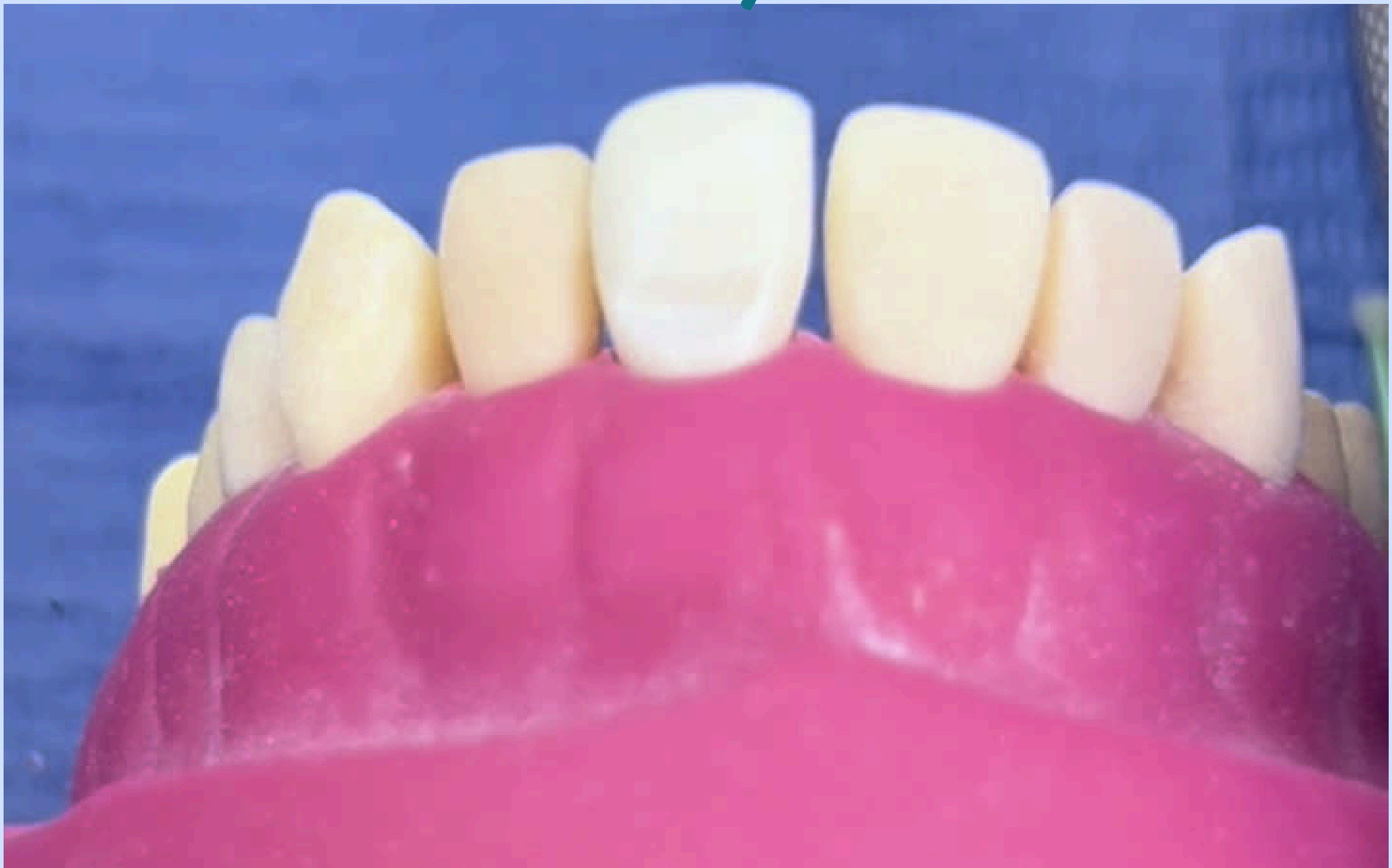
Finishing Discs





Finishing Stones

Class V Cavity Restoration



Finishing your restoration

We promise you after this step, your'e really done!

Let's talk about finishing your restoration.
This step is often skipped by students thinking that it does not matter. Let us tell you why it does matter:

- Removing any roughness on the restoration that could lead to food accumilation if left unfinished.

- Refining the anatomy and contour of the restoration.

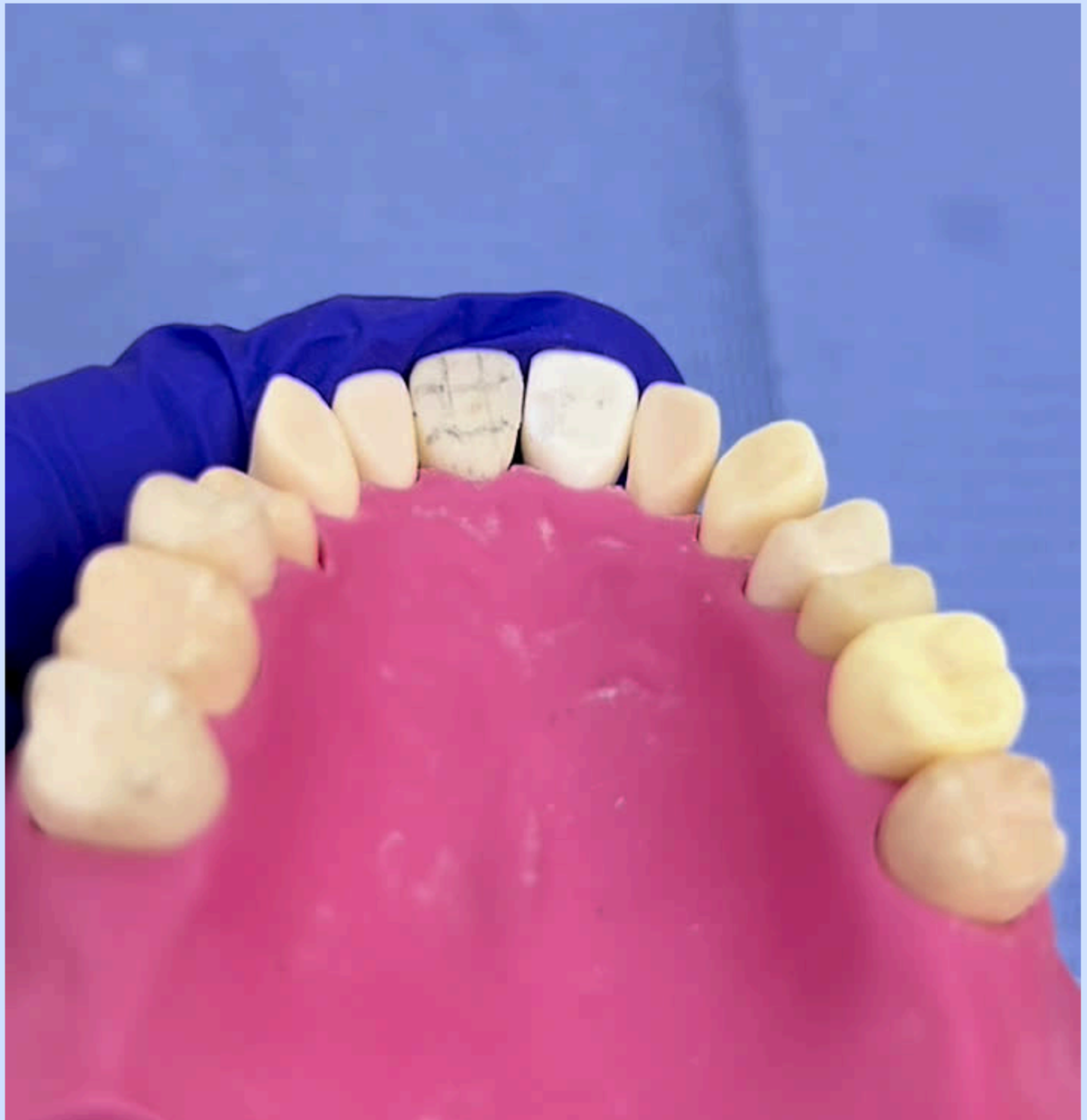
- Remove excess composite.

- Improve esthetics and enhance surface luster.



<u>Conclusion:</u> Do NOT skip finishing if you want that A+ restoration!

Finishing your Restoration



Finishing your Restoration



How to check your work?



Check the MEASUREMENTS done using probe.	Check that there is no undermined enamel at any margin.	check that your restoration is properly finished and polished and there is no roughness.
Check the measurements at different points to make sure the prep is even in all areas.	Check that floor is smooth and even	
check that there is no injury to adjacent tooth (class III)	Using probe, check that your restoration has sealed all margins.	
check that the contact is open properly (class III)	Check that the restoration restored the natural contour of the tooth.	



How to check your work?




Congrats! You have now mastered Class III and V Cavities and Restorations!

Contact Us!

If you have any further questions, stuck in any step or want to show us your progress, feel free to contact us completely free of charge!

Our talented instructor Dr. Salma is waiting for you!

Want to book a one on one session? Just give us a DM!

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THANK
YOU